Mahsa Parsanasab, PhD

Raleigh, NC • mahsa.parsanasab@gmail.com• (949) 774-8772 • LinkedIn

SUMMARY

Data Analyst with 8+ years of experience in computational modeling and data science methodologies, excelling in advanced statistical techniques to derive actionable insights from complex datasets. Proficient in SQL, Python, and MATLAB, with a proven record of enhancing data processing efficiency and accuracy. Committed to applying innovative solutions in industry settings, driving impactful data-driven decisions and advancements.

TECHNICAL SKILLS

- Programming Languages: SQL, Python, R, MATLAB
- Big Data & Distributed Computing: PySpark, AWS S3, Pandas
- Machine Learning & Data Science: NumPy, Scikit-Learn
- Computational Modeling: Monte Carlo Simulations (Command Line Interface)
- Data Visualization: Power BI, Tableau, Matplotlib
- Languages: English (Fluent), Persian (Native), Spanish (Fluent)

EXPERIENCE

Data Analyst

University of California, Irvine | Remote | March 2025 - *Present*

- Improved inverse problem solving accuracy for skin optical property recovery by leveraging adaptive optimization strategies.
- Developed an intelligent iterative Monte Carlo inverse modeling pipeline to recover optical properties from reflectance data. Leveraged a pre-computed Monte Carlo solution database and perturbation method for efficient parameter estimation and optimized convergence.

Graduate Student Researcher

University of California, Irvine | Irvine, CA | September 2019 - March 2025

- Analyzed and managed large-scale unstructured datasets stored in AWS S3, utilizing SQL for data cleaning, transformation, and feature engineering.
- Developed and optimized a PySpark-based data pipeline, achieving a 70% reduction in data processing time.
- Designed and implemented a predictive data model forecasting reflectance measurement using Gradient Boosting.
- Created predictive models for uncertainty quantification in Monte Carlo simulations, improving reliability in multispectral tissue characterization.
- Leveraged clinical data to validate optical imaging technique predictions for diagnosing and monitoring skin diseases, aiding early detection.

Data Analyst

Vista Shimi Kimia | September 2016 - December 2017

- Developed a regression model to forecast raw material consumption by analyzing historical production data via Microsoft Access/SQL, resulting in a 15% reduction in carrying costs and enhanced supply chain efficiency.

EDUCATION

- Ph.D., Chemical and Biomolecular Engineering | University of California, Irvine | 2019 - 2025
- M.Sc., Polymer Engineering | Amirkabir University of Technology | 2014 - 2016
- B.Sc., Polymer Engineering | Amirkabir University of Technology | 2010 - 2014

CERTIFICATIONS

- Data Analytics Professional Certificate, Google | Completed January 2024
- Machine Learning Specialization, DeepLearning.AI | Completed Nov 2022